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VADAPALANI

SMART INTELLIGENT CAUTERY SYSTEM

IN ALLIANCE WITH SDG 3 , 9 & 12

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ABSTRACT



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SMART INTELLIGENT CAUTERY SYSTEM (SICS)



PROBLEM

Conventional cautery systems use fixed power settings, leading to excessive heat, smoke generation, and risk of tissue damage.



OBJECTIVE

To design an intelligent system that automatically regulates cautery power for safer surgical operation.



SOLUTION

A smart, trained cautery control system that adapts power settings based on surgical material input.



CORE METHOD

Closed-loop control based on continuous thermal feedback and real-time monitoring.



NOVELTY

Predictive smoke prevention by reducing power before critical threshold limits are reached.



CAPABILITY

Supports multiple biological materials such as skin, tissue, intestine, and other surgical targets.



OUTCOME

Reduced heat and smoke generation, improved precision, and enhanced surgical safety.

30/1/2026

SMART INTELLIGENT CAUTERY SYSTEM

METHODOLOGIES



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Problem identification

Excess heat and smoke in cautery systems

System design

Define inputs (biological material)

Define outputs (power, alerts, temperature)

Data collection & assumptions

Power ranges

Temperature thresholds

Safe operating zones

Intelligent control logic development

Material-based power selection

Thermal feedback monitoring

Pre-threshold prediction

Power reduction & cut-off logic

Simulation & visualization

Real-time time vs temperature graph

Alerts & status display

Testing & validation (simulated)

Verify response near thresholds

Check stability around safe zone

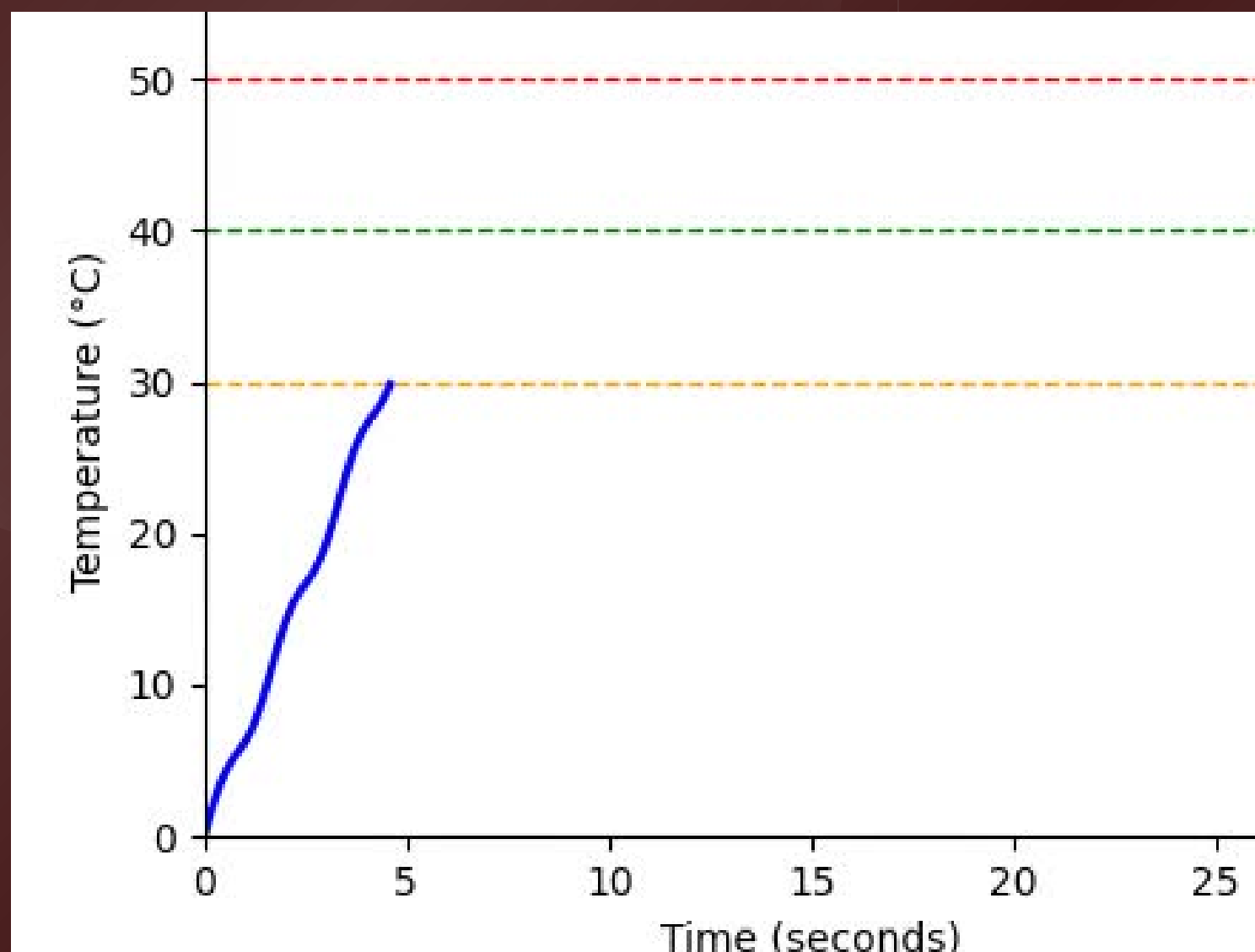
LITERATURE SURVEY



Title of Article	Inference	Author	Journal
Effect of Temperature of an Electrosurgical Unit on Surgical Smoke Distribution: An In-Vitro and Simulation Study.	Establishes the relationship between temperature rise and smoke generation.	Yu, H	PLOS ONE, 2024
Safety Assessment of Electrosurgical Electrodes Using Thermal Injury Evaluation in Organ Tissues	Highlights the need for thermal control to prevent tissue damage.	Zhang, X. R., Wang, Y., et al.	Heliyon, 2024
PID-Fuzzy Control Applied to an Electrosurgical Unit for Output Power Regulation	Presents closed-loop intelligent control techniques for electrosurgical power regulation	Ridha, A. M., Al-Raweshidy, H. S	IEEE Access, 2020
Electrocautery Devices with Feedback Mode and Teflon-Coated Blades Create Less Surgical Smoke for a Quality Improvement in the Operating Theater.	Demonstrates the role of feedback mechanisms in reducing surgical smoke.	Kisch, T., Biel, M., et al.	Medicine (Baltimore), 2015.



Project Presentation



The project is presented using a smart intelligent cautery system in which the user provides the required surgical material input, and the system automatically determines and sets the appropriate power level. Once the procedure begins, the temperature of the cautery tool is continuously monitored and displayed as a real-time time versus temperature graph. When the temperature approaches a predefined pre-threshold value, the system automatically reduces the power to limit further heat rise. As the temperature stabilizes and returns to a lower safe operating zone, the power is increased automatically to maintain effective operation. The entire process is governed by closed-loop control based on thermal feedback, and all control actions are visualized through the real-time graph.



COST ESTIMATION

SMART INTELLIGENT CAUTERY SYSTEM - 3Lakhs (.approx)

Metrics	Approx value	Percentage
Hardware & Core Components	90k	30%
Sensors & Electronics	60k	25%
Software & Intelligent Control	75k	20%
Testing & Validation	45k	15%
Integration & Safety Measures	30k	10%

REFERENCE



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1. Title / Article: Principles and Safety Measures of Electrosurgery
Authors: Ilan Alkatout, Andreas Schollmeyer, Andreas Hawaldar, et al.
Year: 2012

2. Title / Article: Bigger Is Not Always Better: Effects of Electrocautery Settings on Soft Tissue Injury
Authors: Aaron L. Shiver, Jonathan J. S. McDaniel, et al.
Year: 2019

3. Title / Article: Safety Assessment of Electrosurgical Electrodes Using Thermal Injury Evaluation in Organ Tissues
Authors: X. R. Zhang, Y. Wang, Z. Liu, et al.
Year: 2024

4. Title / Article: PID-Fuzzy Control Applied to an Electrosurgical Unit for Output Power Regulation
Authors: A. M. Ridha, H. S. Al-Raweshidy
Year: 2020